DATE:
14/10/2026

Col - AT3

SHORT ANSWER

Q. Definition

TEST

Name:
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S

Regno:
192511416

COURCE: CSA 1516
CODE
COURSE 2: CLOUD
COMPUTING
NAME & BIG *DATA*
ANALYSIS

computing is a fechnology that provider

cloud

computing

Computing

resources

Such an

servers, storage,

databare

fnetworking and software over the internet

without requiring

demand

maintain

physical

Four characteristics :

users

to

630

or

infrastructure.

On demand Self service:

Users

can

access

Computing

resources

whenever

required

without direct.

interaction

with the

Service

Provida.

be

Broad Network Access:

accessed

through

330

cloud

services

can

the internet

using

and tablets

devices such as laptops, smartphones

Resource pooling: Multiple user's share

3

computing resource through

a

common

4

Rapid dasticity: cloud resources

automatically increased

workload requirement

or

decreased

Eg: Amazon Web Services (AWS)

infrastructure.

be

Cer

according

to

allows bussineses

Servers

to rent virtual servers instead of purchasing physical

2) EVOLUTION

OF

CLOUD COMPUTING

Introduction :

(

M

)

ł

cloud
computing
enhancements
in
evolved
through
techno
logies.
Computer hardware,
virtualization
Softw
are
networking & internet
bened

Hardware Evolution

- Main-frame (omputing (1960's):

Large centralized computers

Supported multiple users through

through time

-sharing.

Personal Computers (1980's):

Computing,

accessible to individuals

became

client server Architecture

(1990's): Processing

clients

and

organizations.

and data storage

were

distributed between

& Servers:

could run

on

a

improvin

g

hardware

5

Internet

and

Software

Application:

Virtualization

(2000's):

single

physical

utilization

Multipl

e

virtual

machines

Servering

- Internet expansion High-speed networks

enabled remote

access to computing

resources.

- Web Applications :

Software

became accessible through browsers

without

installation.

- Distributed and Grid computing:

computers worked

- Service

local

and Grid computing: Multiple interconnected together to process large workloads

oriented Architecture :

developed

wing

Applications were

reusable, network

Accessible

drivers.

cloud Services.

Google
cloud

to provide

SERVER

Microsoft Azure,

combined

all

the

a

Scalable

Services

VIRTUALIZATION:

Aws

4

abave
technologies

<>

Server

Virtualization is

cloud

Computing

technology

w that

divider

Server

into

many

independent

a physical virtual

machines

(VM's).

ADVANTAGES

of

SERVER

VIRTUALIZATION:

Since

Resource optimization:

Server

virtualization

divider

Physical Server

-

VM'S

there

is

40

need for more

servers, storage

etc

the

сал

handle everything.

because

2

Costly

Dr

extra

1

the

3

Isolated:

An Server In

the

Virtual

Machines

ane

isolated

from

the

system

environment,

it

does not conflicts

the

Current

running

OS

environment.

No

ent Cost Reduction:

physical

server

infrastructure which

Server Virtualization

all

need of external

voy's

can manage

results in

Cost Reduction.

DATA

DATA

High

Scalability .

of Server

There

are

the

2

Virtualizatio
n

Virtual
computing

advan
tag

داما

cloud
computing

creates abstract It provides

access to computing

C st

over

layer of hardware

Component

s

in order to

Create

environments

Physical

2

efficiency

3

It is

manage

d

machine

It focusses on

hardware

on

a

singl

e

4

multiple virtual

by hypervisor

[a

&

networking

H focus on

delivering

it Scalable software

software which is used
create Virtual
to
environment
s)

Eg: VMWO
ESXsi

Products.

3
R It
It manager
405
automation,
service
managements,
sroit & service interface,
AWS EC2
Resources
Such
al
Server
170
storage,
databa

(2

2

H.

efficient sh

10

hools

157152

DATACENTRE

DATACENTRE

WITH

ADVANT ROLE

EXAMPLE

OF

Datacentre:

Datacentre

1s

a

physica

l

facility

facility

which

consists

Server

Storage

da tabare,

@quipme

nt

and

other

which

&s

used

to

Store

of

networking

network infrastructure.

Process &

manage

the

digital

information.

ROLE OF DATA CENTRE.'

@ Computing

Resources .

High end

High

speed

Server

helps to

execute Software

applications faster

and

loser

requests are

process

ed

quickly & efficiently

Network storage.

Con

High level

2

Storage

Server

bolp

e

cloud servers to connect

can

access

cloud resources

to users, 10

ло

users

(3.

Reliability &

availability:

Backup data

bove Storager,

Back Power

Syrtem

Redis server

setup

efc

can

make

the cloud services

always

available.

4

Data Centre

are

across

different

Security & Discuter

Management: Usually

secured using high level security

firewall

seographical

locations.

Eg:

Google

cloud

has

different data

Centres

across

many

geographical

locations

in

order

to

Serve

seamless

cloud services.

5

CLOUD SERVICE

MODELS

& differentiate

between

IAAS,

PAAS, SAAS,

& XAAS

Aust Cloud

Service Model:

cloud Service

Model

definer

development

platform

how

a

computing Resources

or

Software

Served

to the

users

through the

application is

internet.

7004

cloud Service

Description

Example

Model

99912

IAAS

AWS EC2.

(Infrastructure

on Service)

virtual

110

Server,

© St. Provider Eg:

do computing

resources etc

and entire infrastructure pictz N

you to

architecture

to

the Get Small SAAS /

ot

companies. The

company

should manage

os

& the application

19v157

27910

05121924

@ It provides Eg: Google *App*

12

PAAS
(Platform As Service)

a

3

SAAS
(Software

platform to develop, test and deploy applications without any underlying physical

Cervers

® A full well developed Soth one

an Service application is directly

given to end user

Engine

@ Eg: Google Gmail

260 po

XAAS

(Everything As Service)

4

It provides everythin

g

that

any

virtually

covers

"IT capabilities"

available.

4 Eg: Microsoft Defenda for endpoint.